Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Development of Ultra-Soft and Highly Conductive 3D Printing Resins for High-Performance Flexible Pressure Sensor Platforms**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Chemicals		*Electrical · Electronics*	
	Medium Category	*Small Category*	*Medium Category*	*Small Category*
	Polymeric materials	*Special functional materials technologies*	*Semiconductor devices and systems*	*Sensor element*
Concept of Proposed Technology	**Core Concept** This platform will enable the direct fabrication of microstructures designed for high sensitivity and a wide dynamic range, providing human-skin-like softness while ensuring the electrical reliability required for precise tactile sensing. This project aims to develop the photocurable resin to overcome the limitations in resilience and repulsion forces inherent in conventional Poly(tetramethylene ether) glycol (PTMEG)-based 3D printing resins. Also, this project aims to develop the photocurable resin to overcome the limitations in resilience and repulsion forces inherent in conventional Poly(tetramethylene ether) glycol (PTMEG)-based 3D printing resins.			

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Figure 1. Manufacturing pressure sensors using 3D printing technology

Key Functions and Potential Applications

1) Development of Ultra-soft/Highly Conductive Resin
   - Innovation in Physical Properties: An ultra-low hardness-equivalent to human skin-is achieved, complemented by high resilience that ensures instantaneous recovery even under repeated deformation.
   - Functional Innovation: high electrical conductivity is realized directly through the printing process itself without the need for complex post-processing. Additionally, biocompatibility is ensured by cytotoxicity testing, resulting in a finalized material optimized for wearable devices.
2) Simulation-based Optimization and High-performance Sensor Manufacturing
   - Microstructure Design and Simulation for High-sensitivity/Wide-range Sensing: Both high sensitivity and a wide pressure range—which typically involve a trade-off—are achieved through microstructure design based on Finite Element Analysis (FEA).
   - 3D Printed Pressure Sensor Platform: A high-performance pressure sensor platform is developed by utilizing ultra-soft/conductive materials to print FEA-designed microstructures at high resolution, enabling precise detection across a wide range from subtle micro-pressures to high-pressure levels.

Rationale for Joint R&D Project

Critical Technical Barriers

The primary challenge in developing materials for wearables and robot skins lies in the simultaneous attainment of conflicting physical properties (trade-offs). First is the trade-off between "hardness and elastic recovery". Reducing the hardness of a resin increases hysteresis due to viscoelasticity, which significantly slows the response speed of the sensor [1]. Second is the trade-off between "conductivity and printing suitability". Increasing the conductive filler content raises the viscosity, making precision 3D printing impossible [2]. Realizing both ultra-low hardness and high conductivity by controlling these complex multi-variable correlations is a high-difficulty problem that is challenging to solve through a simple experimental approach alone.

Currently commercialized PTMEG-based resins and conductive paste technologies face technical bottlenecks, as they fail to fundamentally resolve the aforementioned technical challenges such as hysteresis and high viscosity [3].

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Why Global Collaboration?

To overcome these limitations and lead the next-generation wearable pressure sensor market, it is essential for the Korean SME partner to possess specialized resin synthesis capabilities and establish organic collaboration with a Purdue University equipped with advanced simulation technology. In this framework, the Korean SME partner must secure 3D printing resin synthesis and dispersion technologies along with mass-production infrastructure, as seen in global materials companies that export functional resins for LCD/DLP 3D printing to numerous countries.

This joint research aims to transform the existing experience-dependent development paradigm into "Simulation-Driven Engineering" by combining the manufacturing and synthesis expertise of the SME partner with the multiscale simulation technology of Purdue University, which excels in Density Functional Theory (DFT), Ab initio Molecular Dynamics (AIMD), and FEA-based microstructure optimization. Through this integration, the objective is to drastically shorten development timelines, overcome technical bottlenecks, and preoccupy a dominant technological position in the global market.

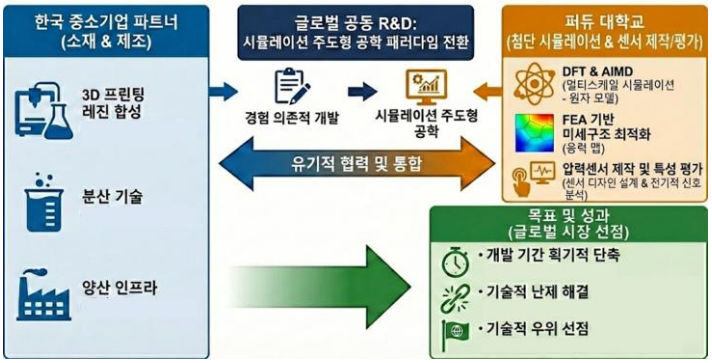

Figure 2. Rationale and objective for global collaborative R&D

Key Words	1. 3D Printing 2. Conductive Resin 3. Pressure sensor 4. Functional microstructure 5. Low Hysteresis

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)

R&D Objectives

R&D Objectives

The primary objective of this project is to develop a next-generation 3D printing flexible pressure sensor material platform that overcomes the limitations of conventional materials (high hardness, low conductivity, and slow recovery). This will be achieved by integrating the "ultra-low hardness/high-resilience polysiloxane synthesis and high-conductivity dispersion technology" of the Korean SME partner with Purdue University's "multiscale simulation and sensor platform technology". Through this collaboration, this project aim to achieve physical properties with a hardness similar to human skin and a conductivity of over 10 S/cm. By commercializing materials that ensure biocompatibility, this project intend to preoccupy the wearable sensor and robot skin markets.

Detailed Objectives

1) Development of Photocurable Polysiloxane Oligomers
2) Synthesis and Development of Siloxane-based Photocurable Crosslinker Monomers
3) Development of Siloxane-based UV Photoinitiator
4) Development of Dispersion Technologies and Formulation for conductive fillers to Achieve 10 S/cm Electrical Conductivity
5) Formulation Development of Ultra-low Hardness, High-resilience, and Conductive Resins Suitable for 3D Printing
6) Microstructure Design and Simulation Optimization for High-sensitivity/Wide-range Pressure Sensing
7) Fabrication and Performance Evaluation of 3D Printing-based Pressure Sensor Platforms

R&D Target Specifications

Category	Parameters	Ratio	Final objective	Year 1	Year 2	Year 3	Year 4	Evaluation method
Resin	Shore Hardness	30 %	10A	40A	30A	20A	10A	ASTM D2240
	Elastic Recovery (%)	20 %	99	-	90	99	99	
	Hysteresis (%)	15 %	< 10	30	15	< 10	< 10	Cyclic Tensile Hysteresis (50% Strain)
	Conductivity (S/cm^2)	10 %	10	-	1	10	10	
	Viscosity (cPs)	5 %	5,000	5,000	5,000	5,000	5,000	ASTM D-1084
	Curing Energy (mJ/cm^2)	5 %	40	-	40	40	40	Energy required for curing 25um
	Cytotoxicity (ISO 10993-5)	5 %	PASS	-	-	-	PASS	ISO 10993-5 2000
Pressure sensor	Sensitivity (kPa^{-1})	40 %	200	140	160	180	200	
	Pressure range (kPa)	30 %	500	200	300	400	500	
	Response/recovery time (μs)	30 %	40/40	160/160	120/120	80/80	40/40	

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R&D Plan	**Year 1 (6 months):** ● Development of low-T_g polysiloxane oligomers ● Initial formulation for 3D printing ● Microstructure design and sensitivity simulation ● Baseline sensor fabrication and testing **Year 2 (12 months):** ● Synthesis of siloxane-based crosslinkers ● Development of conductive resin ● Microstructure design for wide pressure range and high-resolution feature simulation ● Wide-range prototype validation **Year 3 (12 months):** ● Synthesis of High-MW multifunctional oligomers ● Hydrosilylation curing system design ● Multiscale microstructure design for high sensitivity ● Stress distribution simulation and sensor performance evaluation **Year 4 (6 months):** ● Achievement of 10 S/cm conductivity ● Biocompatible hybrid resin finalization ● Fatigue prediction and structural reinforcement ● Final demonstration and reliability testing

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	1) Microstructure Design 2) High-Fidelity Structural Analysis 3) Sensor Fabrication & Performance Validation 4) Reliability & Engineering Guidelines
Expected R&R of Korean SME Partner	1) Material Synthesis 2) Process Optimization 3) Scale-up 4) Commercialization

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Economic and industrial impact** 1) Market Leadership ● Preoccupies the wearable and soft robotics markets by securing a dominant technological edge in functional 3D printing resins. 2) Ecosystem Expansion ● Lowers the barrier for high-performance sensor production, enabling mass customization of complex tactile devices. 3) Global Synergy ● Demonstrates a successful model for Korea-US joint R&D, fostering future collaboration in advanced manufacturing. **Technical Impact** 1) Paradigm Shift ● Transitions development from trial-and-error to “Simulation-Driven Engineering”, drastically cutting R&D time. 2) Trade-off Solution ● Establishes a new material standard by simultaneously achieving ultra-low hardness (<10A) and high conductivity (>10 S/cm). 3) Precision Design ● Enables the design of next-generation sensors with extreme sensitivity and wide dynamic ranges through multiscale simulation. **Social Impact** 1) Healthcare Innovation ● Provides skin-like patches for continuous, non-invasive health monitoring. 2) Enhanced HRI ● Improves Human-Robot Interaction (HRI) safety and capability through the implementation of sophisticated "haptic skins."

6. References

[1] S. Wang et al., “Skin-Inspired Electronics: An Emerging Paradigm”, *Acc. Chem. Res.*, 51(5), 1033-1045 (2018)

[2] A. D. Valentine et al., “Hybrid 3D Printing of Soft Electronics”, *Adv. Mater.*, 29(40), 1703817 (2017)

[3] Shanshan Yao and Yong Zhu, “Wearable multifunctional sensors using printed stretchable conductors made of silver nanowires”, *Nanoscale*, 6, 2345-2352 (2014)